

Numerical Simulation Methods And Their Applications

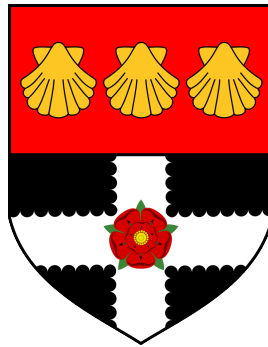
Monte Carlo Methods and Applications in Particle-based Systems

Ridhwaan Amin

32005096

Department of Mathematics and Statistics

University of Reading



April 14, 2026

Plain language summary

This project concerns working with a computer simulation of a particle-based system, particularly using numerical techniques known as Monte Carlo simulations.

It starts with an introduction to the history of these methods, touching upon the Manhattan Project, then focusing on pseudo-random number generators, sampling methodologies, and their applications in statistical mechanics. The project also explores the Metropolis method and its application to a Lennard-Jones liquid. Finally, it concludes with findings of a simulation created from individually developed C++ code.

These findings are

Declaration

I certify that this is my own work and that use of material from other sources has been properly and fully acknowledged in the text. I have read the University's definition of plagiarism and the department's advice on good academic practice. I understand that the consequence of committing plagiarism, if proven and in the absence of mitigating circumstances may include failure in the Year or Part or removal from the membership of the University. I also certify that neither this piece of work, nor any part of it, has been submitted in connection with another assessment.

In submitting this work I can confirm that I ~~have~~/have not used Generative AI Tools to prepare and complete my assignment.

to prepare and complete work, please one select the following statements:

~~a./I acknowledge the use of [AI] to generate materials to support my research and understanding on the topic when drafting this assignment, but none of the outputs are included in the final version./~~

~~b./I acknowledge the use of [AI] to generate materials that were included within my final assessment in modified form./I have taken responsibility to ensure that where generated materials are included, these have been referenced appropriately, in accordance with the requirements of the assignment brief./~~

Contents

List of Tables	v
List of Figures	v
1 Development Of Computer Simulation Methods	1
1.1 The Manhattan Project	1
1.1.1 Background	1
1.1.2 Stanisław Ulam	1
1.1.3 Subsequent Developments	1
2 Monte Carlo Methods	2
2.1 Pseudo-Random Number Generators	2
2.1.1 Linear Congruential Generator (LCG)	2
2.1.2 Mersenne Twister	2
2.2 Sampling Methodology	2
2.2.1 Simple	2
2.2.2 Importance	2
2.3 Statistical Mechanics	3
2.4 The Metropolis Method	3
3 The Lennard-Jones Potential	4
3.1 A Description	4
3.1.1 Sketching The Potential	4
3.1.2 Applications in Modelling Simple Liquids	4
3.2 Periodic Boundary Conditions	4
3.2.1 Figure	4
4 Practical Application in a Numerical Simulation	5
4.1 Code and Development	5
4.1.1 Steps Toward Implementation	5
4.1.2 Code Origin and Inspiration	5
4.2 Application to a Lennard-Jones liquid	5
4.2.1 Numerical Analysis	5
5 Conclusions, Findings, Prospectives	6
6 Bibliography	7

A Monte Carlo C++ Code	8
B Approximation of Pi in Python, Using Monte Carlo Techniques	9

List of Tables

List of Figures

1 Development Of Computer Simulation Methods

The first concepts of Monte Carlo simulations can be found in the 18th century, contained within a report found in the 1733 edition of 'Histoire de L'Académie Royale des Sciences'. This report - created by Georges Louis Leclerc, Comte de Buffon - makes mention of an approximation of π by throwing needles or rods atop a floor of boards, measuring the fractions that fell on a single board. Of course, this can be recreated in the modern day by a simulation of approximating π using many points (see Appendix B). When initially proposed in this report, the field of Monte Carlo methods had not been distinctly named or partitioned. (Landau & Binder 2021)

1.1 The Manhattan Project

1.1.1 Background

Before understanding Monte Carlo Methods, and the higher level category of 'Computer Simulations', we are bound to mention **The Manhattan Project**. The top-secret research and development program was the first to produce Nuclear Weapons, and was a key playing card in the context of the greater geopolitical climate: World War II.

Funded by nations such as the UK, Canada and, majorly, the USA, the Manhattan Project is often cited as a turning point in human history. Disregarding moral arguments concerning atomic weaponry, a large factor in the awe of the project was the pure advancement in scientific knowledge - building upon previous discoveries of fission.

1.1.2 Stanisław Ulam

Stanisław Ulam was prominent in contributing to the founding of Monte Carlo Methods as a designated methodology. Famously, Ulam thought up the concept while recovering in a hospital from surgery. He had been playing solitaire to pass the time and wondered whether the probability of success could be estimated statistically. Rather than approaching the problem from a combinatorial approach (which he had already tried and found unsuccessful) he began to think more 'abstractly' - he had realised that he could simply observe successive games and note outcomes. Crucially for Stanisław Ulam, this could be extended to many simulated trials with the recent advent of computers - which would yield increasingly accurate estimates. (Eckhardt 1987)

1.1.3 Subsequent Developments

(Summerscales 2023)

2 Monte Carlo Methods

2.1 Pseudo-Random Number Generators

Random number generation is an important part of Monte Carlo methods. There are multiple different methods to obtaining random numbers, but the algorithms we are focusing on within this report are called **Pseudo-Random Number Generators** (PRNGs). While they do not generate 'actual' random numbers, what they generate is sufficient enough in variability that it can be used in scenarios where random numbers are needed.

The way that they achieve this is through the use of mathematical formulae, varying in levels of complexity. The algorithms can get increasingly elaborate, yet the simplest can consist of only a few lines of code utilising basic mathematical arithmetic. Sequences are generated which attempt to approximate random numbers, based on an initial starting *seed* (Gentle 2003).

While humans can not predict the next number in the sequences that are generated, this is not perfectly computationally secure. This is because if the initial *seed state* is known, an identical sequence can be reproduced. Therefore, the sequences are deterministic and not as unpredictable as systems with true chaos built into them - such as radioactive atom decay, or, famously, *Cloudflare's* wall of lava lamps (Liebow-Feeser 2017).

2.1.1 Linear Congruential Generator (LCG)

One such example, albeit a simple one, is the Linear Congruential Generator given in the following form:

$$x_i \equiv (ax_{i-1} + c) \bmod m, \quad \text{with } 0 \leq x_i < m;$$

2.1.2 Mersenne Twister

The Mersenne Twister is a more complex, modern PRNG developed in 1997 (Matsumoto & Nishimura 1998).

It derives its name from a prime number known as a Mersenne Prime, which contributes to its period. As its period is sufficiently large, the algorithm is highly optimised, and the numbers generated have low correlations, the Mersenne Twister is a prominent PRNG that is used across industry. Most notably, it is used in scientific simulations (Monte Carlo methods, as discussed) but also within Machine Learning, statistical sampling and also video game graphics (Samadov 2024).

2.2 Sampling Methodology

2.2.1 Simple

2.2.2 Importance

$$I = \int_a^b f(x) dx$$

2.3 Statistical Mechanics

2.4 The Metropolis Method

3 The Lennard-Jones Potential

3.1 A Description

The Lennard-Jones potential describes the potential energy of two non-bonded particles based on their distance of separation. The equation accounts for both attractive forces and repulsive forces.

The equation is given as follows

$$V(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

where:

- V indicates the intermolecular potential between particles,
- ε is the well depth: the maximum strength of attractive force and minimum energy value
- σ

3.1.1 Sketching The Potential

3.1.2 Applications in Modelling Simple Liquids

3.2 Periodic Boundary Conditions

3.2.1 Figure

4 Practical Application in a Numerical Simulation

4.1 Code and Development

4.1.1 Steps Toward Implementation

4.1.2 Code Origin and Inspiration

My code was built on a framework of **Fortran** code from 'Understanding Molecular Simulation' by Daan Frenkel and Berend Smit (Frenkel & Smit 1996). Another Monte Carlo simulation of the Lennard-Jones model (Kelter et al. 2015), packaged within a library known as NetLogo (Wilensky 1999), also provided much guidance and inspiration. These were adapted and rewritten from scratch, by me, into **C++**. My program can be found in the appendix of this report (see Appendix A).

4.2 Application to a Lennard-Jones liquid

4.2.1 Numerical Analysis

5 Conclusions, Findings, Prospectives

6 Bibliography

- Eckhardt, R. (1987), 'Stan Ulam, John von Neumann, and the Monte Carlo Method', *Los Alamos Science* (15), 131–141. Also: Los Alamos National Laboratory Tech. Report LA-UR-88-9068.
URL: <https://permalink.lanl.gov/object/tr?what=info:lanl-repo/lareport/LA-UR-88-9068>
- Frenkel, D. & Smit, B. (1996), *Understanding Molecular Simulation: From Algorithms to Applications*, Academic Press.
- Gentle, J. E. (2003), *Random Number Generation and Monte Carlo Methods*, second edn, Springer.
- Kelter, J., Luijten, E. & Wilensky, U. (2015), 'Netlogo monte carlo lennard-jones model', <http://ccl.northwestern.edu/netlogo/models/MonteCarloLennard-Jones>.
- Kopka, H. & Daly, P. W. (1999), *A Guide to LaTeX*, third edn, Pearson Education.
- Landau, D. P. & Binder, K. (2021), *A Guide to Monte Carlo Simulations in Statistical Physics*, fifth edn, Cambridge University Press.
- Liebow-Feeser, J. (2017), 'Randomness 101: Lavarand in production', <https://blog.cloudflare.com/randomness-101-lavarand-in-production/>.
- Matsumoto, M. & Nishimura, T. (1998), 'Mersenne twister: a 623-dimensionally equidistributed uniform pseudo-random number generator', *ACM Trans. Model. Comput. Simul.* **8**(1), 3–30.
URL: <https://doi.org/10.1145/272991.272995>
- Metcalfe, T. (2023), 'What was the manhattan project?', *Scientific American*.
- Samadov, I. (2024), 'The mersenne twister algorithm. a deep dive into a pseudo-random number generator', *Medium*.
- Sharma, V. (2023), 'Estimating pi using monte carlo methods', <https://medium.com/the-modern-scientist/estimating-pi-using-monte-carlo-methods-dbf26c888d6>.
- Summerscales, O. (2023), 'Hitting the jackpot: The birth of the monte carlo method', <https://www.lanl.gov/media/publications/actinide-research-quarterly/1123-hitting-the-jackpot-the-birth-of-the-monte-carlo-method>.
- Wang, C. (2018), *Applications of Monte Carlo Methods in Studying Polymer Dynamics*, PhD thesis, University of Reading.
- Wilensky, U. (1999), 'Netlogo', <http://ccl.northwestern.edu/netlogo/>.

A Monte Carlo C++ Code

ParticleSimulation/src/main.cpp

```
#include <iostream>

int initialisation() {

    // Particle Placement

    return 0
}

void mcmove() {

    // Particle Displacement

}

int main() {

    // Metropolis Algorithm

    double Temp;
    int NumParticles;
    double Density;
    int Cycles;

    initialisation();

    return 0;
}
```

B Approximation of Pi in Python, Using Monte Carlo Techniques

ApproximationOfPi/src/python/SimpleImplementation.py

```
#!/ python3
"""
Simple implementation of approximation of pi.

Steps:
Generate random x, y inside centre of square of side 2
Repeat steps for a large number of points
Calculate ratio of points
Multiply by 4 (pi / 4)
"""

import numpy as np

def estimatePi(samples):
    insideCircle = 0
    for i in range(samples):
        x = np.random.uniform(-1, 1)
        y = np.random.uniform(-1, 1)

        if (x**2 + y**2) <= 1:
            insideCircle += 1
    ratio = insideCircle / samples
    return (ratio * 4)

if __name__ == "__main__":
    print(f"Estimate with 1 Sample:\n{estimatePi(1)}\n")
    print(f"Estimate with 10 Sample:\n{estimatePi(10)}\n")
    print(f"Estimate with 50 Sample:\n{estimatePi(50)}\n")
    print(f"Estimate with 100 Sample:\n{estimatePi(100)}\n")
    print(f"Estimate with 1000 Sample:\n{estimatePi(1_000)}\n")
    print(f"Estimate with 10,000 Sample:\n{estimatePi(10_000)}\n")
    print(f"Estimate with 1,000,000 Sample:\n{estimatePi(1_000_000)}\n")
    print(f"Estimate with 50,000,000 Sample:\n{estimatePi(50_000_000)}\n")
```

ApproximationOfPi/src/python/GraphicalRepresentation.py

```
#!/ python3
"""
Graphical representation of approximation of Pi using Monte Carlo methods.
"""
import matplotlib.pyplot as plt
import numpy as np

from SimpleImplementation import estimatePi

def main():
    samples = int(input("Enter Intended Samples:\n"))

    x = np.random.uniform(-1, 1, samples)
    y = np.random.uniform(-1, 1, samples)

    plt.plot(x, y, ".k", markersize = 1)

    t = np.linspace(0, 2 * np.pi, 400)
    xc = np.cos(t)
    yc = np.sin(t)
    plt.plot(xc, yc, 'r-', linewidth=1)
    plt.gca().set_aspect('equal', adjustable='box')

    plt.xlim(-1.1, 1.1)
    plt.ylim(-1.1, 1.1)
    plt.xlabel("x")
    plt.ylabel("y")

    plt.suptitle(f"Estimate of Pi: {estimatePi(samples)}", fontsize = 18)
    plt.title("Graphical Representation Is Independent To Estimate Calculation",
              fontsize = 10)
    plt.show()

if __name__ == "__main__":
    main()
```